

No. of Functional Features Included in the Solution

Date	3 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

#	Feature Name	Description	Location
1	Model Comparison Dashboard	Home page displays accuracy and ROC-AUC for all 4 trained models (LR, DT, RF, XGBoost) in a comparison table, highlighting Random Forest as the best model	templates/index.html + app.py / route
2	15-Field Applicant Prediction Form	Input form captures: income type, income amount, education, family status, housing type, occupation, age, employment years, children count, family members, gender, car/property ownership, and 4 contact flags	templates/predict.html
3	Instant Pre	After form submission, displays Approved (green) or Rejected (red) result card	templates/predict.html result section

#	Feature Name	Description	Location
	dicti on Res ult	with animated confidence progress bar and probability percentage	
4	Pre dicti on Hist ory	Stores and displays the last 10 predictions made in the current browser session, with collapsible detail cards showing all input values	templates/history.html + Flask session
5	Scenari o Pre-Fill Car ds	Three buttons on the dashboard (Low Risk / Mid Risk / High Risk) that automatically fill the prediction form with representative applicant profiles for demo purposes	templates/index.html + static/js/script.js
6	Real - Time Inco me For mat ter	As the user types annual income, a display label shows the formatted value (e.g., "₹4,80,000 / year") in real time using JavaScript	static/js/script.js updateIncomeDisplay()
7	Em ploy men t Stat us Aut o-Tog gle	When the user selects "Unemployed" as income type, the employment years field automatically resets to 0 and IS_UNEMPLOYED flag is set to 1	static/js/script.js updateUnemployed()
8	Res pon sive Ban king	Bootstrap 5 responsive grid ensures the app works on mobile, tablet, and desktop. Navy + gold banking color theme with Font Awesome	static/css/style.css (--primary-blue, --accent-gold)

#	Feature Name	Description	Location
	UI	icons	
9	History Clear	Button on the History page sends POST to /clear_history, resets Flask session, and redirects to empty history — allows starting fresh	templates/history.html + app.py /clear_history route
10	Graceful Error Handling	If model PKL files are missing (e.g., first clone without training), MODELS_LOADED=False prevents crashes and shows a user-friendly error message	app.py MODELS_LOADED flag